

Requirements-based testing of neural networks.

A critical review of RBT4DNN, its empirical evidence, and two focused additional analyses.

SEMINAR

AI Testing Landscape: Ensuring Robustness through Advanced Testing Methods

COURSE LEADERSHIP

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PRESENTATION

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Three questions guide the argument.

01 • NEED

Why do neural-
network tests need
requirements?

PROBLEM + BACKGROUND

02 • METHOD

How does RBT4DNN
generate and evaluate
targeted tests?

APPROACH + METHODOLOGY + RESULTS

03 • EVIDENCE

How convincing—and
scalable—is the
resulting evidence?

CRITIQUE + ADDITIONAL ANALYSES

High average accuracy does not answer a specific requirement.

CONVENTIONAL TESTING Known behavior A specification maps an input condition to an expected output. Test oracles can check the result directly.	DNN TESTING Learned behavior The model is inferred from examples. Global accuracy summarizes a distribution, not every meaningful scenario.	THE MISSING LINK Semantic scenarios Developers need evidence for statements such as “a very thick 7 must still be classified as 7.”
GLOBAL METRIC How often is the complete test set correct?	→ SEMANTIC SLICE Which rare inputs does the score contain?	→ REQUIREMENT VERDICT Does the model satisfy this specific contract?

[1] §1-§2.5 · “model under test” means the evaluated neural network

Existing generators rarely start from the requirement itself.

Neuron coverage, image transformations, and search-based generators can produce many tests—without proving that those tests represent the scenario a stakeholder cares about.

INPUT	Requirement	Structured natural language defines a semantic precondition and an expected outcome.
METHOD	RBT4DNN	Fine-tune a generator on images that satisfy that precondition.
OUTPUT	Targeted tests	Run the learned component and evaluate the requirement-specific oracle.

[1] §1, §2.5 and §3 - authors claim the first requirement-driven DNN test-input generator

Four domains test whether one workflow survives increasingly ambiguous semantics.

MNIST · 7 REQUIREMENTS MNIST artifact samples Source-linked in interactive deck MNIST Measured height, thickness, and lean. A controlled starting point. Output: digit class	CELEBA-HQ · 7 REQUIREMENTS CelebA-HQ artifact samples Source-linked in interactive deck CelebA-HQ Ambiguous attributes labeled by visual question answering or humans. Output: eyeglasses decision	SGSM · 7 REQUIREMENTS SGSM artifact samples Source-linked in interactive deck SGSM Synthetic driving scenes with spatial relations linked to steering and acceleration. Output: numeric control	IMAGENET · 4 REQUIREMENTS ImageNet artifact samples Source-linked in interactive deck ImageNet Animal morphology linked to allowed classes in the WordNet taxonomy. Output: class set
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[1] Table 2 and §4.1.1-§4.1.3 · samples linked from the pinned public artifact

MNIST M3 artifact samples
Source-linked in interactive deck

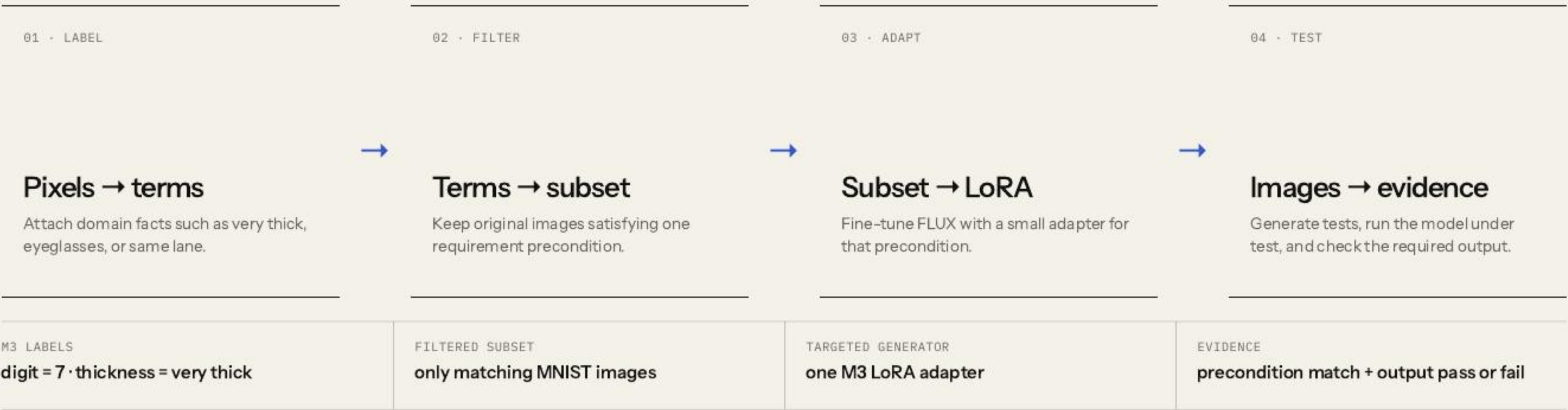
A requirement defines both relevance and correctness.

If the digit is a 7 and is very thick, then the model shall label it as 7.

PRECONDITION	POSTCONDITION
Which images count as relevant tests?	Which model outputs count as acceptable?

[1] Table 2, M3 · a finite passing suite cannot prove the universal requirement

RBT4DNN turns one requirement into a targeted test campaign.



[1] Fig. 2 and §3 · symbols are translated into an explicit input/output contract

Glossary terms make visual requirements operational.

GLOSSARY TERM

A named visual property that can be assigned to an image: “very thick,” “same lane,” or “eyeglasses.”

01 · DEFINE

Authors specify the glossary

Requirement concepts are manually turned into a finite vocabulary of visual properties.

TERM CREATION IS NOT AUTOMATIC

02 · LABEL ORIGINAL DATA

Labels come from domain-specific signals

MNIST measurements · SGSM scene graphs · MiniCPM visual Q&A for CelebA-HQ and ImageNet.

CELEBA ALSO HAS HUMAN LABELS

03 · USE THE LABELS

Filter first; check later

Labels select the LoRA training subset. Separate binary classifiers then check glossary terms on generated images.

PRECONDITION MATCH · RQ1 / RQ3

Boundary: separate training labels and evaluation classifiers reduce direct reuse, but neither constitutes independent semantic ground truth.

[1] §3.2 and §4.1.6 · mean glossary-classifier accuracy 94.5%; C4 is 70.8%

One filtered precondition subset trains one targeted LoRA.



BENEFIT	Generate many new candidates inside a rare requirement region
EVIDENCE SOUGHT	Relevance, distributional plausibility, and semantic variation
NOT ESTABLISHED	Non-memorization or systematic coverage within that region

[1] §3.3 and §4.1.4 - LoRA equation simplified; latent-space coverage remains future work

The six research questions validate inputs before interpreting model outcomes.

RQ = Research Question. The numbered requirements themselves are M1–M7, C1–C7, S1–S7, and I1–I4.

Validate the generated inputs RQ1–RQ3

RQ1	Consistency	Does the image satisfy the precondition?
RQ2	Plausibility	Is its distribution close to matching training images?
RQ3	Variation	Are other glossary features distributed similarly?

Evaluate the resulting tests RQ4–RQ6

RQ4	Comparison	Are targeted outcomes more informative than baselines?
RQ5	Violations	Do tests reveal credible postcondition failures?
RQ6	Diagnosis	Do failures expose repeatable decision patterns?

[1] §4.1 · input validity is established before model outcomes are interpreted

Targeted LoRAs satisfy the requested precondition 92.1% of the time on average.

92.1%

mean across 21 requirements in MNIST, CelebA-HQ, and SGSM.

M3 · targeted LoRA

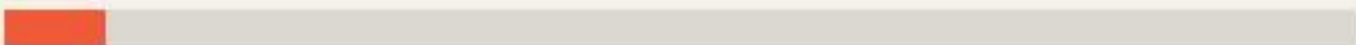
filtered images: very thick 7



95.8%

M3 · one all-data LoRA

all annotated MNIST training images; not plain FLUX



7.4%

COMPARISON

Precondition-specific LoRA vs one LoRA trained on all annotated data, plus DeepHyperion and transformations · 100 images × 10.

BOUNDARY

Glossary classifiers average 94.5% accuracy; they are measurement proxies, not semantic ground truth.

[1] §4.1.4, §4.2.1 and Fig. 3 · prompt-only FLUX is not part of RQ1

Targeted LoRA is much closer for MNIST and SGSM; CelebA-HQ gains are smaller and mixed.

77.09%

reported average KID reduction across Table 3's MNIST M1-M7 and SGSM S1-S7 rows.

COMPARE	targeted LoRA ↔ prompt-only FLUX
SAME INPUT	identical precondition prompt
CELEBA-HQ	12.9% automated · 27% human labels

KID · KERNEL INCEPTION DISTANCE

A distance between two image sets in a learned feature space · lower means the distributions are closer.

BOUNDARY

KID is not an image-level realism probability; several reference subsets contain fewer than 300 images.

[1] §4.2.2 and Table 3 · paper prose says M1-M6, but 77.09% matches M1-M7 arithmetic

Targeted LoRA stays closer to the filtered feature mix in all three domains.

MNIST · MEAN JS 7.2× closer PROMPT-ONLY FLUX TARGETED LORA .1765 .0244	CELEBA-HQ · MEAN JS 1.4× closer PROMPT-ONLY FLUX TARGETED LORA .1150 .0831	SGSM · MEAN JS 5.5× closer PROMPT-ONLY FLUX TARGETED LORA .1960 .0355
JS · JENSEN-SHANNON DIVERGENCE Compare frequencies of the other glossary terms with the matching training subset · lower means closer.		BOUNDARY Means derived from Table 4; JS does not measure pixel diversity, memorization, or latent-space coverage.

[1] §4.2.3 and Table 4 · displayed domain means are calculated from the seven reported rows

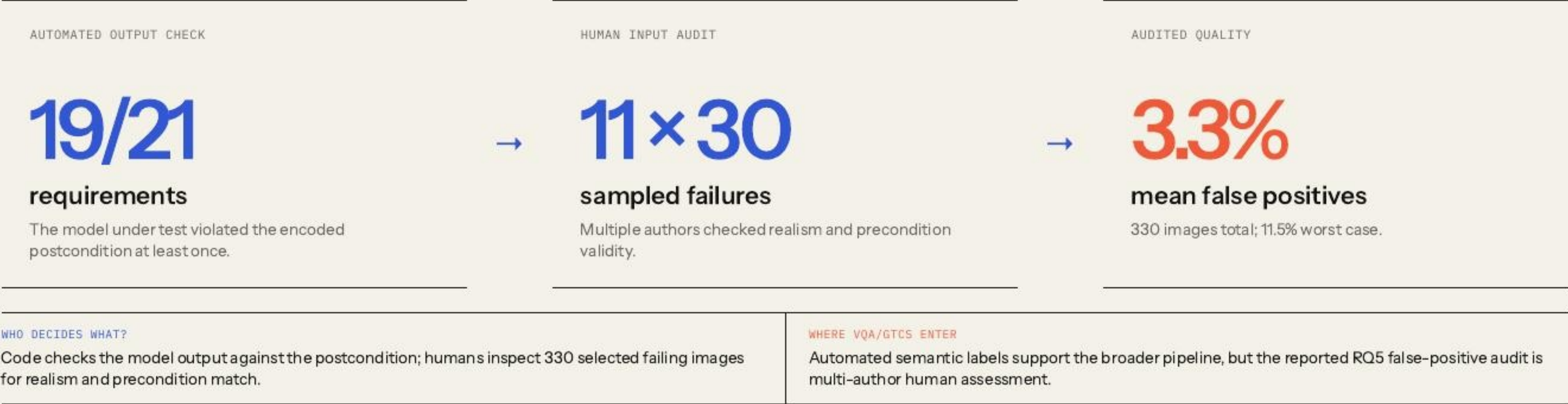
High pass rates are uninformative when generated images miss the requirement.

Technique	Passed Tests	Failed Tests
Deephyperion		
Image Transform		
RBT4DNN		

Original paper evidence · Table 5 · cropped only for legibility · CC BY 4.0.

DEEPHYPERION Searches a user-defined feature map for diverse, boundary-region tests; it is not conditioned on M2's "very thick" term.	INTERPRETATION RULE Both baseline rows mostly miss thickness; only after input relevance is established does pass/fail address M2.
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RBT4DNN exposed failures broadly, and the audited subset was mostly valid.



[1] §4.1.6 and §4.3.2 · assessor disagreement on precondition consistency was 1.7%

Backgrounds and partial animals align with repeated wrong labels—but do not prove causation.

Failure Cases with Model Output

Worm Fence Lycaenid Daisy Sandbar

Sussex Spaniel Wallaby Worm Fence White Wolf

Barn Spider Tick Lemon Centipede

Train Data

Original paper Table 7 · generated failure cases and training examples · cropped for legibility · CC BY 4.0

METHOD

4 requirements × 3 classifiers × repeated generation

Collect 1,486 postcondition failures, rank frequent wrong labels, then manually inspect associated images.

EXAMPLE

A bird near a fence → “worm fence”

Other cases align with insects, horses, or snakes that resemble frequent training classes or backgrounds.

BOUNDARY

Association, not causal diagnosis

The study does not change only the background while holding the animal fixed.

The paper demonstrates feasibility—not a complete assurance process.

GENERATIVE TRAINING Better tests Improve generator training and adaptation to reveal more observed failures and reduce false positives.	COVERAGE EVIDENCE Latent-space coverage Map the model's internal feature space and deliberately search regions that generated tests have not exercised.	GENERALIZATION More domains Expand driving-like safety datasets and explore broader robustness and functional requirements.
ASSURANCE BOUNDARY Automated labels, selected datasets, narrow baselines, and small KID subsets limit how far the evidence generalizes.		

[1] §4.4 and §5 · author-stated scope kept separate from this seminar's critique

The paper jointly validates 330 sampled failures—but not the full failure population.

PRECONDITION INVALID · MODEL PASSES Irrelevant pass The output says nothing about the stated requirement.	PRECONDITION INVALID · MODEL FAILS Off-target false positive A model error is not yet a requirement counterexample.
PRECONDITION VALID · MODEL PASSES Valid requirement pass Relevant evidence, but never proof of the universal property.	PRECONDITION VALID · MODEL FAILS Credible counterexample This joint event is the paper’s strongest testing evidence.

SAMPLED JOINT AUDIT

30 failures × 11 low-pass requirements = 330 independently assessed samples.

Most defensible critique: semantic validity is both the method’s key input and its main measurement risk. The 330-image human audit is a real strength; the remaining gap is independent, blinded, population-wide joint validation.

Three lightweight checks test reproducibility and prioritize stronger human validation.

01 · ARTIFACT CONSISTENCY

Recalculate stored MNIST results

Compare seven published fields with the authors' saved result files. No model inference, generation, or retraining.

max Δ · .013

02 · AUDIT PRIORITY

Combine match and failure rates

Compute $P(\text{match}) \times P(\text{fail})$ as an independence-based triage score—not an observed joint probability.

S2 · 73.6%

03 · EXTERNAL SPOT-CHECK

Ask one multimodal model

Sample two images per requirement and judge only precondition validity: 42 images in total.

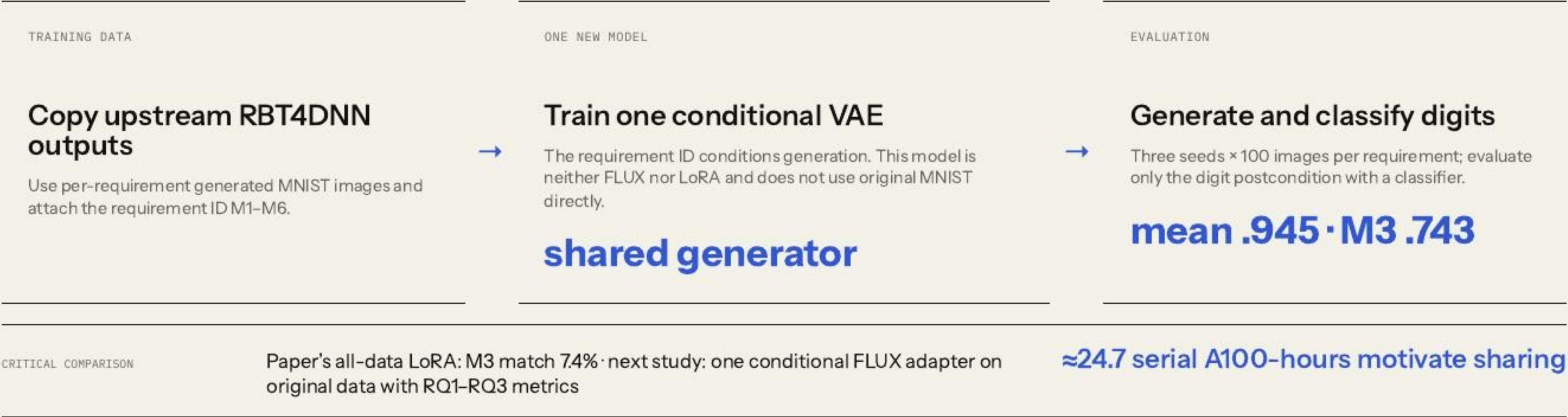
31/42 · .738

WHAT THIS ESTABLISHES

The stored numbers are internally consistent and some requirements deserve priority review. It does not estimate the population-wide valid-failure rate.

One conditional VAE matched digit-pass rates—but did not test requirement validity.

VAE = Variational Autoencoder: an encoder compresses images into a probabilistic latent space; a decoder samples that space to generate images.



[1] Fig. 3 · seminar mnist-shared-generator · no direct precondition, KID, JS, or human-validity evaluation

Requirements make tests **relevant**. Validity makes failures **credible**.

WHAT THE PAPER ESTABLISHES

Structured requirements can guide distributionally plausible, glossary-diverse, scenario-specific test generation across four domains.

WHAT LIMITS THE CLAIM

Semantic labels and glossary classifiers are useful proxies; credible failures still depend on joint input-and-output validation.

MOST VALUABLE NEXT STUDY

Compare conditional shared and per-requirement adapters with identical RQ1-RQ3 metrics and independent human audits.

References.

[1]

N. J. Mozumder, F. Toledo, S. Dola, and M. B. Dwyer. “RBT4DNN: Requirements-based Testing of Neural Networks.” arXiv:2504.02737, v3, 2025. CC BY 4.0.

[2]

E. J. Hu et al. “LoRA: Low-Rank Adaptation of Large Language Models.” ICLR, 2022.

[3]

RBT4DNN artifact. github.com/less-lab-uva/RBT4DNN · seminar provenance pinned to commit `c3deff871ed41043d83a05289faab84c41706586`.

[4]

Seminar experiments. Reproducible notebooks, results, summaries, and provenance in this repository.

Original paper ↗	Original artifact ↗
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